MANCHESTER 1824
The University of Manchester

RISK REGISTER FOR 3rd YEAR PROJECT

Project Title:	**A Real-Time Haptic Rendering System Integrating Physics-Based Simulation and OpenGL Visualisation**		Submission Date:	1/05/2026	
Student Name:	11349510				

Project Risk	Severity			Potential			Score (Severity x Potential)	Mitigation Measures
	L	M	H	L	M	H	L=1, M=2, H=3	
Electrical faults during wiring/testing (short circuit, incorrect connections)	X				X		2	Use low-voltage systems; power off before modifying wiring; check connections before powering; use insulated connectors
Injury from moving mechanical parts (rotating joints, links)		X			X		4	Limit actuator torque; avoid contact during operation; test at low power; implement watchdog shutdown
Actuator overheating during prolonged operation		X		X			2	Monitor device temperature; limit runtime; allow cooling periods; avoid sustained high torque
Mechanical failure (belt slip, loose components)		X			X		4	Secure all components; inspect before testing; avoid excessive loading; operate within tested limits
Control instability causing unexpected motion or oscillations			X		X		6	Use conservative control gains; include damping; implement torque limits and rate limiting; test incrementally
Communication failure (loss of command stream)		X			X		4	Implement watchdog timeout to zero torque; validate communication before operation
Software bugs causing incorrect behaviour		X			X		4	Incremental testing; logging and debugging; isolate subsystems during development
Component failure (e.g. actuator failure during project)		X			X		4	Design for partial operation; maintain flexibility in testing; prioritise core functionality
Use of tools during assembly (cuts, slips)	X				X		2	Use appropriate tools; follow workshop safety procedures; maintain tidy workspace
Extended computer use (eye strain, fatigue)	X					X	3	Take regular breaks; maintain ergonomic posture; adjust screen setup